

Signalling part 3

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Microeconomics 2 Spring 2025

Readings: MSPC 5.4

Overview

In this lecture we look more formally at two signalling problems:

- A labor market model in a simplified setting with numbers and discrete education choice (no graphs)
- An example from corporate finance.

Education and signalling

Players: A worker and two firms.

Timing:

- Nature chooses ability $\theta \in \{\theta_L = 2, \theta_H = 5.5\}$.
 - $Pr(\theta = 2) = 0.5$
- Worker chooses education level, $e \in \{0, 1\}$. Bemærk: Kun to mulige niveauer.
- Firms/employers each offer a wage contract $w(e)$.
- The worker either accepts a contract (randomizes if both firms offer the same) or rejects both
- Output equals θ [unobserved and non-contractible]

Payoffs:

$$\pi_{worker} = \begin{cases} w - 8\frac{e}{\theta} & \text{if a contract is accepted} \\ 0 & \text{if both contracts are rejected} \end{cases}$$

$$\pi_{employer} = \begin{cases} \theta - w & \text{if contract is accepted} \\ 0 & \text{if contract is rejected} \end{cases}$$

We look for equilibria

- Signalling game $\Rightarrow$ pooling and separating eqa are possible
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 - 2 potential pooling eqa:

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- Signalling game $\Rightarrow$ pooling and separating eqa are possible
 - 2 potential separating eqa:
 $(e_H^*, e_L^*) = (1, 0)$ and $(e_H^*, e_L^*) = (0, 1)$.
 - 2 potential pooling eqa:
 $e_H^* = e_L^* = 1$ and $e_H^* = e_L^* = 0$.

Perfect Bayesian EQ

Perfect Bayesian equilibrium is a strategy profile σ^* and a system of beliefs μ^* satisfying Requirements 1-3.

Requirement 1: At each information set, the player with the move must have a belief about which node in the information set has been reached by the play of the game. For a non-singleton information set, a belief is a probability distribution over the nodes in the information set; for a singleton set, the player's belief puts probability one on the single decision node.

Perfect Bayesian EQ

Requirement 2: Given their beliefs, the players' strategies must be sequentially rational. That is, at each information set the action taken by the player with the move (and the player's subsequent strategy) must be optimal given the player's belief at the information set and the other player's subsequent strategy (where a subsequent strategy is a complete plan of action covering every contingency that might arise after the given information set has been reached).

Requirement 3: The players' beliefs in every information set (on and off the equilibrium path) are determined by the players' equilibrium strategies and Bayes' rule whenever applicable.

Pooling equilibria

What must hold for a pooling equilibrium

- For education: Both types choose same education level e^* .
- Both types are paid the same wage, $0.5 \cdot 2 + 0.5 \cdot 5.5 = 3.75$
- One way to find eqa is to go through potential possibilities.

Pooling equilibria

Consider the following pooling eqm:

$$e^*(\theta_L) = e^*(\theta_H) = 0$$

$$\mu(0) = Pr(\theta = 5.5 | e = 0) = 0.5$$

$$w(0) = w(1) = 3.75$$

$$\mu(1) = Pr(\theta = 5.5 | e = 1) = 0.5$$

- First set of beliefs are on eqm path, second are off eqm path.
- This is an eqm because:
 - Given the strategies of firms, the strategy of the worker is optimal (requirement 2). The firms' beliefs secure that workers get same wage for both $e = 0$ and $e = 1$. No reason to deviate.
 - Given the worker's strategy, firms' beliefs on the eq path are set appropriately (requirement 3)

Pooling equilibria

Another potential pooling eqm:

$$e^*(\theta_L) = e^*(\theta_H) = 1$$

But this is not incentive compatible, as low productivity type would deviate even if out of eqm beliefs have $Pr(\theta_L|e=0) = 1$ (i.e. worst possible beliefs) and thus wages equal to 2:

$$U_L(e=0) = 2 > U_L(e=1) = 3.75 - \frac{8}{2} = -0.25$$

Thus this potential pooling eqm fails requirement 2.

Separating equilibria

What must hold for a separating eqm?

- For education: $e^*(\theta_H) \neq e^*(\theta_L)$
- For beliefs: $Pr(\theta = \theta_i | e = e^*(\theta_i)) = 1$
- For wages: $w^*(e^*(\theta_H)) = 5.5$ and $w^*(e^*(\theta_L)) = 2$
- Consider the following candidate:

$$e^*(\theta = 2) = 0, \quad e^*(\theta = 5.5) = 1$$

$$w^*(e = 0) = 2, \quad w^*(e = 1) = 5.5$$

$$Pr(\theta = 5.5 | e = 0) = 0$$

$$Pr(\theta = 5.5 | e = 1) = 1$$

Separating equilibria

We now show that the candidate separating eq is indeed a PBE using the standard constraints studied so far in this course. Contracts must maximize the utility of θ_L and θ_H subject to two sets of constraints (PCs and ICs):

- Note: firms should not make losses (they could simply not offer contracts instead)
 - $w(0) \leq \theta_L = 2$
 - $w(1) \leq \theta_H = 5.5$
 - Competition btw employers drive employers to make zero profit. So hold with equality.
- PC worker? Worker always has possibility of $e = 0$ and getting positive payoff.

Separating equilibria

- IC: self selection constraints. For separating eqm we need contracts to be incentive compatible
 - $U_L(e = 0) \geq U_L(e = 1)$ and $U_H(e = 1) \geq U_H(e = 0)$
or
 $w(0) - 0 \geq w(1) - \frac{8}{2}$ and $w(1) - \frac{8}{5.5} \geq w(0) - 0$
or in eqm above:
 $2 \geq 1.5$ and $5.5 - 1.4545 \geq 2$
- Beliefs:
 - On eqm path?
 - Off eqm path?

Separating equilibria

Hence the separating eqm satisfies:

- Requirement 1: beliefs are formed at all info sets
- Requirement 2: worker's strategy is optimal given firms' strategies
- Requirement 3: the firms' beliefs on the eqm path is determined by the worker's eqm strategy

Other potential separating eqm

$$e^*(\theta_L) = 1$$

$$e^*(\theta_H) = 0$$

$$w(1) = 2$$

$$w(0) = 5.5$$

$$Pr(\theta = 2|e = 0) = 0$$

$$Pr(\theta = 2|e = 1) = 1$$

Not incentive compatible.

Link to screening

- When we were talking about screening in a similar setup we showed that there could be a pooling contract which breaks the separating eqm if $E(\theta)$ was high enough to make the pooling contract more attractive than the costly action (see screening slides).
- BUT this is not relevant for signalling because of the timing of the game: employers do not compete by offering contracts but by responding to signals!
- This is thus another example of self-fulfilling beliefs, and illustrates how powerfull beliefs are in PBE.

Equilibrium refinements

The pooling eq does not survive the eqm refinements we have studied.

- θ_L has a payoff of 3.75 in eqm.
- i.e. compared to eqm payoff the low type would never be incentivized to deviate regardless of beliefs, as even with highest possible wage that could be optimal for the firms, the low type's utility would be $5.5 - \frac{8}{2} = 1.5 \leq 3.75$.
- So eqm payoff dominates the best which could happen in a deviation for the low type.
- Did we use the dominance or the intuitive criterion?

Efficiency

- The separating equilibrium is constrained efficient (or second best efficient).
- A social planner with same informational constraints cannot improve the allocation.
- A law banning education will not work as you make high type worse off. He prefers signalling: $w(1) - \frac{8}{2} = 4.05 > 3.75$.
- It is not first best efficient (or generally efficient) as a social planner who observes types would not need education.

Signalling in financial markets (Credit markets)

Foundation of modern corporate finance:

Modigliani-Miller theorem: "The form and source of financing is irrelevant for firms' investment decisions."

"The basic theorem states that in the absence of taxes, bankruptcy costs, agency costs, and asymmetric information, and in an efficient market, the value of a firm is unaffected by how that firm is financed. Since the value of the firm depends neither on its dividend policy nor its decision to raise capital by issuing stock or selling debt, the Modigliani–Miller theorem is often called the capital structure irrelevance principle."

Empirically we observe that new equity issues on average result in a drop in the stock price of the issuer. This is somewhat paradoxical, since a new issue is usually made because the firm has new investment opportunities.

- In a frictionless capital market, firms maximize profits and undertake only positive NPV projects (NPV: net present value. Positive NPV $\Rightarrow$ return outweighs all costs)
- What happens to MM if we introduce asymmetric information? Can equity issue be a signal? And of what?
- We illustrate this with a stylized example.

Setup

- Players: Firms and investors. All are risk neutral.
- Assume there are many potential investors so they only require that their initial investment is covered (no positive expected profits to investors)
- Firm has private information about the value of its assets
- The firm's decision to raise new capital acts as a signal.

Firms

- Current assets: Return is 1 with prob. γ_i , 0 with prob. $1 - \gamma_i$.
- New project: Return is 1 with prob. η_i , 0 with prob. $1 - \eta_i$.
Requires investment $I = 0.5$.
- Two types of firms: G or B , $pr(G) = pr(B) = 0.5$. It holds that $\gamma_G \geq \gamma_B$ and $\eta_G \geq \eta_B$.
- Only the manager of the firm observes the true state of nature.
- If manager has 0.5 in cash when are new projects taken?

Financial contract

Assume $\eta_B \geq 0.5$ (i.e. in both the good and bad state of the world the probability of succes is above 0.5)

- i.e it is always efficient to undertake the project.

A financial contract (written to raise capital to invest in the new project) can be specified by the ex post repayment to investors.

- Gross value of firm: $V \in \{0, 1, 2\}$
- Let r_i denote the repayment when the gross value is i .
- If both projects fail the value is 0 and the firm cannot make any repayment, therefore the contract focus on r_1 and r_2 (i.e. repayment under 1, 2).
- Note that investors will generally just require expected repayment to equal initial investment (risk neutral).

Equity financing

Issuing equity implies that the buyer becomes a partial owner of the firm. Therefore it will hold that the repayments will be the same share of revenue and thus $2r_1 = r_2$.

Under this constraint, the firm faces a very simple signalling problem with only two actions:

- 1 Issue and invest equity
- 2 Do not issue

Type B firms

At worst, investors believe the firm is bad and invest 0.5 in the firm for a share s of the firm. Expected profits are given by

$$E(\text{profits}) = Pr(\text{success}) * (\text{share of profits}) * \text{return} = (\gamma_B + \eta_B) * s * 1$$

The share is determined such that investors' expected return equals their investment:

$$(\gamma_B + \eta_B) * s * 1 = 0.5 \Leftrightarrow s = \left[\frac{0.5}{\gamma_B + \eta_B} \right]$$

This share implies that the firm gets a payoff equal to

$$\eta_B \left[1 - \frac{0.5}{\gamma_B + \eta_B} \right] + \gamma_B \left[1 - \frac{0.5}{\gamma_B + \eta_B} \right] = \gamma_B + \eta_B - 0.5$$

Does the firm undertake the project in the worst case? Is this better than without the project? What are the implications?

Type G firms

If type G firms also undertake the project then we arrive at a pooling eqm where the investors' share is given by:

$$0.5(\gamma_B + \eta_B) * s * 1 + 0.5(\gamma_G + \eta_G) * s * 1$$

$$\Rightarrow s = \frac{0.5}{0.5(\gamma_B + \eta_B) + 0.5(\gamma_G + \eta_G)}$$

Compared to first best

$$\frac{0.5}{\gamma_G + \eta_G} < \frac{0.5}{0.5(\gamma_B + \eta_B) + 0.5(\gamma_G + \eta_G)} < \frac{0.5}{\gamma_B + \eta_B}$$

This implies that bad firms are better off and good firms are worse off than if they could credibly tell investors they are good firms.

A type G firms obtain the following expected payoff if it issues equity:

$$\begin{aligned}
 & (\gamma_G + \eta_G) \left[1 - \frac{0.5}{0.5(\gamma_B + \eta_B) + 0.5(\gamma_G + \eta_G)} \right] \\
 &= \gamma_G + \eta_G - \frac{\gamma_G + \eta_G}{(\gamma_B + \eta_B) + (\gamma_G + \eta_G)} - 0.5 + 0.5 \\
 &= \gamma_G + \eta_G - 0.5 - \frac{(\gamma_G + \eta_G) - 0.5(\gamma_B + \eta_B + \gamma_G + \eta_G)}{(\gamma_B + \eta_B) + (\gamma_G + \eta_G)},
 \end{aligned}$$

where last term is positive.

Does the manager issue equity?

Pooling equilibrium

$$\eta_G \geq 0.5 + \frac{(\gamma_G + \eta_G) - 0.5(\gamma_B + \eta_B + \gamma_G + \eta_G)}{(\gamma_B + \eta_B) + (\gamma_G + \eta_G)} \quad (1)$$

a pooling eqm can be supported (both firms will invest given eqm path beliefs which are 0.5).

In (1) the “gain from investing” exceeds the extra cost associated with issuing equity

Dilution cost - based on pooling belief

- The last term in (1) reflects the subsidy from a type G firm to the investors (to cover for the potential that they are dealing with a type B firm)
- It is the cost of incomplete information
- When

$$\eta_G < 0.5 + \frac{(\gamma_G + \eta_G) - 0.5(\gamma_B + \eta_B + \gamma_G + \eta_G)}{(\gamma_B + \eta_B) + (\gamma_G + \eta_G)} \quad (2)$$

the manager of good firms will not issue equity, and only firms in the bad state issue. In this case the firms action perfectly reveal (or signal) the firm's information and we have a separating equilibrium.

- In (2) the “gain from investing” is below the extra cost associated with issuing equity.
- Note that the smaller γ_B, η_B the more likely (2) is to hold.
- bad quality drives out good

Separating equilibrium - Comments

- The value of the firm is lower than it would have been if he had sufficient funds to invest
- For good firm, positive NPV project are not undertaken
- When the bad firm announces new equity, investors learn that the firm is of type B ; as a result the firm value in the market falls from

$$0.5\gamma_G + 0.5(\gamma_B + \eta_B - 0.5)$$

to

$$\gamma_B + \eta_B - 0.5$$

Equilibrium

- Case 1: The separating eqm exists and a pooling eqm cannot exist because the dilution cost (based on pooling) is so high that G type will not issue equity
- So even if investors expect ex ante that both types issue equity it is not worthwhile for high type to invest.
- Of course in eqm investors know this and expect G -type firms to not invest and this is fulfilled

Equilibrium

- Case 2: The separating eqm exists because the investors believe only type B invest! (if they expected pooling G firms would prefer to pool and the pooling eqm would prevail)
- The eqm “survives” because investors expect separation ex ante and this implies that it is optimal for G firms NOT to pool
- self-fulfilling beliefs.

Equilibrium

- Case 3: The pooling eqm exists because dilution cost (dilution cost based on pooling) is so low that G type will issue equity
- Off eqm path beliefs can be arbitrary (both types prefer investment to none regardless)
- self-fulfilling beliefs.
- the condition in (1) is satisfied